H5 - Zhussupbekova Zere IT-2401

a) $z_{LP} \leq z^*$ in three lines. Every integer-feasible point $y \in \{0;1\}^3$ of the triangle ILP also satisfies $0 \leq y \leq 1$ (since $\{0;1\} \subset [0;1]$) and the same three covering rows - so it is also LP-feasible. Hence the ILP's feasible set is a subset of the LPs feasible set, over the same objective $c^T y$. Minimizing the same function over a larger set can only give a value that is less than or equal to the value over the smaller set, so $z_{LP} = \min_{LP\text{-feasible}} c^T y \leq \min_{ILP\text{-feasible}} c^T y = z^*$

b) three gaps, using $z_{LP} = 8$

point	value z	$(z - z_{LP})/z$	role
true integer optimum (1;1;0)	$z^* = 10$	$(10-8)/10 = 0.2$ (20%)	quality of LP bound
feasible point (1;0;1)	11	$(11-8)/11 = 0.273$ (27.3%)	quality of this solution
rounded-up point (1;1;1)	16	$(16-8)/16 = 0.5$ (50%)	quality of this solution

c) Which number tells you what

The first number (using the true optimum z^*) is an intrinsic property of the LP relaxation itself - how loose z_{LP} is as a lower bound on the best possible integer answer, independent of any particular algorithm. The other two (using a merely-feasible z, not necessarily optimal) measure the quality of that specific solution relative to the LP bound - a practical proxy for „how close to optimal is what my heuristic found" which is exactly what we will compute in check point 4 where the true z^* of the real (much larger) problem is unknown and only z_{LP} (from here) and a heuristic's z are available. A large value of that second kind of gap could mean either a weak heuristic or a loose LP bound - it alone cannot tell you which, only comparing several heuristic against the same z_{LP} strats to.